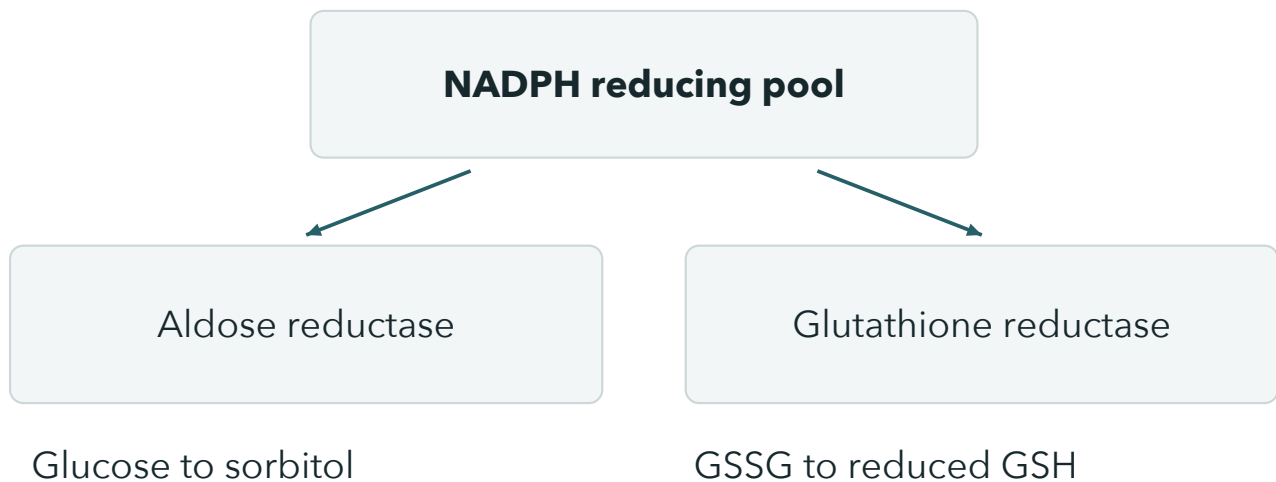


Lens redox and protein homeostasis

Part 1 | Two demands on a shared cofactor

The polyol pathway can impose oxidative as well as osmotic stress.

Competing uses; not identical consequences



NADPH supports antioxidant recycling and is also consumed by aldose-reductase activity.

Separate the consequences

Sorbitol accumulation promotes osmotic stress; increased NADPH consumption can constrain regeneration of reduced glutathione. These are related but distinct hypotheses. Measure cell volume and antioxidant redox state separately instead of inferring both mechanisms from lens opacity alone.

Trace the cofactor before the product

NADPH and NADH have different roles in this comparison.

Reaction

Feature	Cofactor direction	Interpretation
Glucose to sorbitol	NADPH consumed	Potential reducing-capacity competition
Sorbitol to fructose	NAD ⁺ reduced to NADH	Different cofactor pool
Oxidised to reduced glutathione	NADPH consumed	Antioxidant regeneration

A controlled rescue

Restoring antioxidant capacity without normalising polyol accumulation would help distinguish an oxidative component from the persistent osmotic component. A single nonspecific improvement does not establish which pathway was limiting; use orthogonal measurements to constrain the inference.

Crystallins require ordered organisation

Gross lens shape can remain stable while microscopic protein organisation becomes optically abnormal.

Level

Feature	Change	Optical consequence
Molecular	Oxidation and aggregation	Index inhomogeneity
Microstructural	Uneven protein organisation	Light scatter
Gross geometry	Unchanged curvature	No obligatory specific paraxial-power change

Separate outcomes

Scatter and refractive power are not interchangeable measures.

Opacity can reduce contrast even when a nominal prescription is unchanged.

Protein abundance alone does not prove protein organisation is preserved.

Chaperone action differs from synthesis

A protein can limit aggregation without making more lens proteins.

A chaperone experiment

- Apply a controlled aggregation stress
↓
- Hold substrate protein amount constant
↓
- Compare soluble and aggregated fractions
↓
- Test addition of alpha-crystallin

What rescue establishes

Improved solubility supports an anti-aggregation effect under the experiment's conditions. It does not prove reversal of every established cataract or unlimited repair capacity. Compare protein organisation and optical scatter alongside a biochemical assay rather than treating all three as identical outcomes.

Choose a measurement matching the mechanism

Cataract is a clinical endpoint with multiple upstream processes.

Hypothesis

Feature	Useful measurement	Why
Osmotic stress	Cell volume/polyol content	Tests solute-associated swelling
Redox stress	Glutathione redox state	Tests antioxidant balance
Aggregation	Soluble protein/scatter	Tests organisation and optical effect

Reasoning rehearsal

If two treatments improve opacity equally, compare their effects on each mechanism-specific readout. One may improve hydration and another protein stability. Equal final appearance is weaker evidence of a shared mechanism than a matched sequence of intermediate biological changes.